BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE
MID-SEMESTER EXAMINATION (I[ST] SEMESTER) 2016-2017
TRANSPORT PHENOMENA (CLOSE BOOK)

CE F212
Max. Marks: 60

Dated: 7.10.2016
Max. Duration: 90 minutes

Answer all questions. Begin each answer on a new page. Any assumptions can be made reasonably, if needed.

1. Write only the option number for i to vi in your answer book. (5 x 6 = 30)

i. What pressure gradient along the streamline, *dp/ds,* is required to accelerate water in a horizontal pipe at a rate of 30 m/s^2?

a) 20.0 kPa/m
b) 30.0 kPa/m
c) 3.0 kPa/m
d) -30.0 kPa/m

ii. Consider a compressible fluid for which the pressure and density are related by $p/\rho^n = C_o$, where n and C_o are constants. Integrate the equation of motion along the streamline, to obtain the "Bernoulli equation" for this compressible flow which is given as

a) $\frac{n-1}{n}\frac{p}{\rho}+\frac{v^2}{2}+gz = cons\tan t$

b) $\frac{n}{n-1}\frac{p}{\rho}+\frac{v^2}{2}+gz = cons\tan t$

c) $\left(\frac{n}{n-1}-1\right)\frac{p}{\rho}+\frac{v^2}{2}+gz = cons\tan t$

d) $\frac{p}{\rho}+\frac{v^2}{2}+gz = cons\tan t$

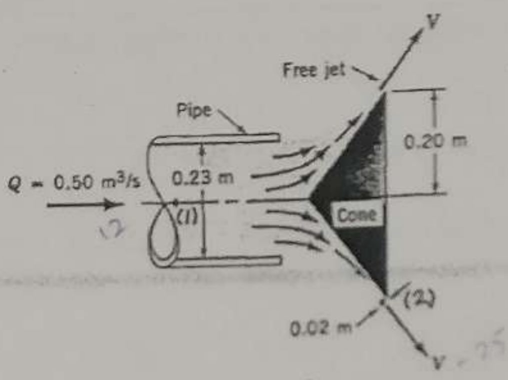
iii. Small-diameter, high-pressure liquid jets can be used to cut various materials. If viscous effects are negligible, estimate the pressure needed to produce a 0.1 mm diameter water jet with a speed of 700 *m/s.* The flowrate is

a) 5.45×10^5 kN/m^2; 2.5×10^{-5} m^3/s
b) 4.25×10^5 kN/m^2; 5.5×10^{-3} m^3/s
c) 2.45×10^5 kN/m^2; 5.5×10^{-6} m^3/s
d) 4.45×10^5 kN/m^2; 5.5×10^{-6} m^3/s

An open tank has a vertical partition and on one side contains gasoline with a density = 700 kg/m^3 at a depth of 4 m. A rectangular gate that is 4 m high and 2 m wide and hinged at one end is located in the partition. Water (density = 1000, g = 9.81) is slowly added to the empty side of the tank. The depth, *h,* of water at which the gate starts to open is

a) 3.55 m
b) 2.50 m
c) 5.55 m
d) 5.33 m

v. A conical plug is used to regulate the air flow from the pipe as shown in Fig. The air leaves the edge of the cone with a uniform thickness of 0.02 m. Make necessary assumption. If viscous effects are negligible and the flowrate is 0.50 m^3/s, then the pressure within the pipe is

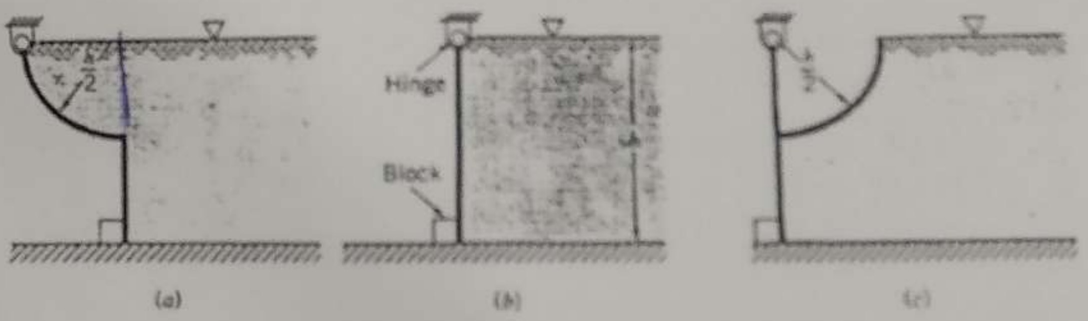


a) 515 N/m^2
b) 105 N/m^2
c) 255 N/m^2
d) 155 N/m^2

vi. A closed, 0.4 m-diameter cylindrical tank is completely filled with oil ($SG = 0.9$) and rotates about its vertical longitudinal axis with an angular velocity of 40 rad/s. Determine the difference in pressure just under the vessel cover between a point on the circumference and a point on the axis.

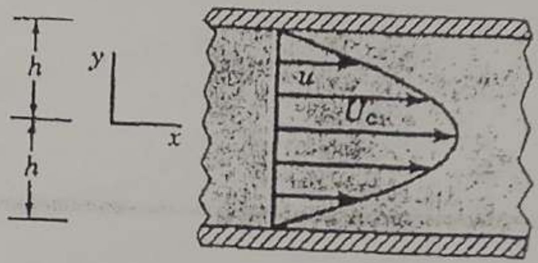
a) 288.0 kPa
b) 28.8 kPa
c) 20.8 kPa
d) 208.0 kPa

2. Three gates of negligible weight are used to hold back water in a channel of width h as shown in Fig. The force of the gate against the block for gate (b) is R. Determine (in terms of R) the force against the blocks for the other two gates. (15)



3. The center pivot irrigation system is to provide uniform watering of the entire circular field. Water flows through the common supply pipe and out through 10 evenly spaced nozzles. Water from each nozzle is to cover a strip 30 feet wide as indicated. If viscous effects are negligible, determine the diameter of each nozzle, d_i, $i = 1$ to 10, in terms of the diameter, d_{10}, of the nozzle at the outer end of the arm. (8)

4. It is known that the velocity distribution for two-dimensional flow of a viscous fluid between wide parallel plates is parabolic; that is $u = U_c\left[1-\left(\frac{y}{h}\right)^2\right]$ and $v=0$. Determine, if possible, the corresponding stream function and velocity potential. Give proper explanation. (7)



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BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE
COMPREHENSIVE EXAMINATION IST SEMESTER 2016-2017
TRANSPORT PHENOMENA

CE F212
Max. Marks: 45

Dated: 12.12.2016
Max. Duration: 90 minutes

1. ***Answer all questions.***
2. ***Begin each answer on a new page.***
3. ***Assume data reasonably, if needed.***
4. ***Underline your final answer for each question***

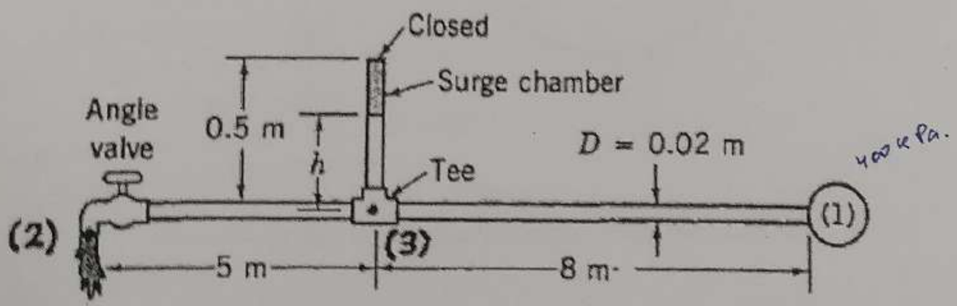
PART B (OPEN BOOK) - 45 MARKS

8. A spherical tank of diameter *D* has a drain hole of diameter *d* at its bottom. A vent at the top of the tank maintains atmospheric pressure within the tank. The flow is quasi-steady and inviscid and the tank is full of water initially. Take g=32.2 ft/s^2.
   i. Determine the water depth as a function of time, $h = h(t)$.
   ii. Tabulate 11 data points at equal interval (h vs t) for $0 \leq h \leq 20$
   iii. Plot graph of $h(t)$ *vs* t for tank diameter (D) of 20 *ft*, if $d = 1$ in.
   iv. Also find the expression for the time to empty the tank and its value for the given data in (iii).

(8+2+3+2=15)

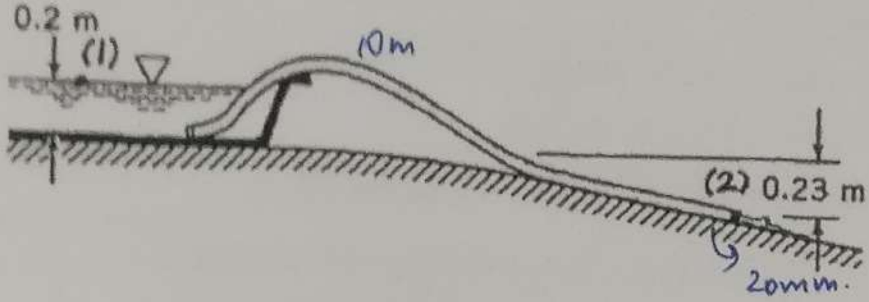
9. When the valve is closed the pressure throughout the horizontal pipe shown in Figure is 400 kPa, and the water level in the closed surge chamber is $h = 0.4$ m. Assume the friction factor is $f = 0.02$ and the fittings are threaded fittings (K_L=2 for angle valve and K_L=0.9 for tee). If the valve is fully opened and the pressure at point (1) remains 400 kPa,
   i. Find the velocity at (2).
   ii. Find P_3 in both the cases.
   iii. Determine the new level of the water (h) in the surge chamber.

(2+3+5=10)



0.A smooth plastic, 10-m-long garden hose with an inside diameter of 20 mm is used to drain a wading pool as is shown in Figure below.

i. If viscous effects are neglected, what is the flowrate from the pool?
ii. Also find flowrate, if viscous effects are included.
iii. What is the percent increase or decrease in flowrate in (ii) compared to (i)?



11. A fan (see Figure) has a bladed rotor of 12-in.-outside diameter and 5-in.-insi and runs at 1725 rpm. The width of each rotor blade is 1 in. from blade inlet to volume flowrate is steady at 230 ft^3/min and the absolute velocity of the air at bla is purely radial. The blade discharge angle is 30° measured with respect to th direction at the outside diameter of the rotor. Take $\rho = 2.38 \times 10^{-3}$ slugs/ft^3.

i. Draw the velocity triangles at inlet section (1) and at outlet section (2)
ii. What would be a reasonable blade inlet angle (measured with respect to th direction at the inside diameter of the rotor)?
iii. Find out mass flow rate at outlet section (2)
iv. Find U_2, $W_{\theta,2}$, $V_{\theta,2}$ (the symbols have usual meaning as explained in provided in lecture slides).

(3+